GroupDrive

Product Requirements Document · built on the Google Drive API v3

Field	Value
Status	Draft v1.0
Date	2 November 2026
Platform	Web app (desktop and mobile browsers)
Core dependency	Google Drive API v3 (+ Google Picker API)

1. Summary

GroupDrive is a workspace for student group projects that sits on top of Google Drive. One person creates a project; GroupDrive creates a shared Drive folder, invites teammates, turns open document comments into a to-do list, shows who changed what, and on deadline day exports the final work to PDF and freezes the submitted version. Students keep working in the Google Docs, Sheets and Slides they already use. GroupDrive adds the coordination layer that Drive is missing.

2. Problem

• Group project files end up scattered across several people's Drives, with inconsistent sharing settings.

• Feedback lives in document comments, so nobody has one list of what is still open.

• Students can't easily see who has contributed, which causes conflict and makes fair grading hard.

• At the deadline, someone has to export the right version to PDF by hand, and later edits can overwrite what was actually submitted.

3. Goals and non-goals

Goals

• Set up a shared, correctly permissioned project folder in under 1 minute.

• Give every team one list of open tasks, generated from Drive comments.

• Show a simple, factual contribution history per teammate.

• Make submission one click: export to PDF and keep that exact version forever.

• Request only the non-sensitive `drive.file` scope, so no restricted-scope security assessment is needed.

Non-goals (v1)

• Replacing Google Docs editing, chat apps, or a learning management system (Canvas, Moodle, etc.).

• Grading, plagiarism detection, or judging the quality of contributions.

• Access to a user's whole Drive. GroupDrive only sees files it created or that the user picked.

• Native mobile apps.

4. Users

Persona	Does	Needs
Student project lead	Sets up the project, invites the team, makes the final submission.	Wants to stop chasing people and files.
Student teammate	Edits documents and resolves comments.	Wants to know what's left to do and get credit for their work.
Instructor / TA (read-only, P2)	Views a project's submitted PDFs and contribution summary.	Wants fair evidence when a team reports a problem.

5. User stories

ID	Story	Priority
US-1	As a lead, I create a project with a name, a deadline and teammates' emails, so everyone gets the same folder with the right access.	P0
US-2	As a teammate, I create a new Doc/Sheet/Slides file from GroupDrive, so it lands in the project folder automatically.	P0
US-3	As a teammate, I add an existing file from my Drive with a file picker, so old work joins the project.	P0
US-4	As a teammate, I see all unresolved comments from every project file as one to-do list, and can resolve them from there.	P0
US-5	As a lead, I click Submit to export chosen files to PDF into a Submissions folder and lock the submitted versions.	P0
US-6	As a teammate, I see an activity feed of recent edits, so I know what changed since I last looked.	P1
US-7	As any member, I see a contribution summary (revisions and resolved comments per person).	P1
US-8	As a lead, I can switch everyone to view-only after the deadline so the work can't change.	P2
US-9	As an instructor, I open a read-only link to a team's submissions and summary.	P2

6. Functional requirements and Drive API mapping

Each requirement names the Drive API v3 call that implements it. Paths are relative to `https://www.googleapis.com`.

Requirement	API call	Notes
FR-1 Create project folder (US-1)	`POST /drive/v3/files`	mimeType `application/vnd.google-apps.folder`; also create a *Submissions* subfolder.
FR-2 Invite teammates (US-1)	`POST /drive/v3/files/{fileId}/permissions`	type=user, role=writer on the project folder; Drive sends the invite email.
FR-3 Create new file in project (US-2)	`POST /drive/v3/files`	Google Docs / Sheets / Slides mimeType, parents=[projectFolderId].
FR-4 Add existing file (US-3)	`Google Picker, then PATCH /drive/v3/files/{fileId}`	Picker grants drive.file access; addParents=projectFolderId.
FR-5 List project files	`GET /drive/v3/files`	q="'{folderId}' in parents and trashed=false".
FR-6 Unresolved comments to-do list (US-4)	`GET /drive/v3/files/{fileId}/comments`	fields include resolved, author, quotedFileContent; show where resolved=false.
FR-7 Resolve comment from app (US-4)	`POST /drive/v3/files/{fileId}/comments/{commentId}/replies`	Reply with action=resolve and optional note.
FR-8 Activity feed (US-6)	`GET /drive/v3/changes/startPageToken, GET /drive/v3/changes, POST /drive/v3/changes/watch`	Webhook channel for near-real-time updates; poll as a fallback. Stop with POST /drive/v3/channels/stop.

Requirement	API call	Notes
FR-9 Contribution summary (US-7)	`GET /drive/v3/files/{fileId}/revisions`	Count revisions by lastModifyingUser; combine with FR-6/FR-7 comment data.
FR-10 Export to PDF (US-5)	`GET /drive/v3/files/{fileId}/export`	mimeType=application/pdf; upload result to Submissions via POST /upload/drive/v3/files.
FR-11 Freeze submitted version (US-5)	`PATCH /drive/v3/files/{fileId}/revisions/{revisionId}`	keepForever=true on the head revision; store revisionId with submission record.
FR-12 Lock after deadline (US-8)	`PATCH /drive/v3/files/{fileId}/permissions/{permissionId}`	Change role writer to reader for teammates; reversible by the lead.
FR-13 Instructor access (US-9)	`POST /drive/v3/files/{fileId}/permissions`	role=reader on Submissions folder only.
FR-14 Storage check	`GET /drive/v3/about`	fields=storageQuota,user; warn before exports if the lead's Drive is nearly full.

7. Permissions, privacy and security

• **Scope:** request only `https://www.googleapis.com/auth/drive.file` (non-sensitive). This avoids the restricted-scope verification and security assessment that full `drive` access requires.

• **Consequence of drive.file:** GroupDrive can only see files it created or that a user opened with it (e.g. via Picker). Files a teammate creates directly in the Drive UI won't appear. The app must push users to create files through GroupDrive (FR-3) or add them with the Picker (FR-4), and show a clear hint in the UI.

• Files stay in users' own Google Drive. GroupDrive stores only IDs, project metadata, comment/revision counts and submission records.

• OAuth tokens are encrypted at rest. Users can disconnect at any time, which deletes their tokens and cached metadata.

• No document content is stored on GroupDrive servers except the moment an export is streamed back into Drive.

8. Non-functional requirements

• **Performance:** project dashboard loads in under 2 s for projects with up to 50 files.

• **Freshness:** activity feed and to-do list reflect Drive changes within 1 minute (push), or 5 minutes (polling fallback).

• **Quota safety:** request only needed `fields`, batch where possible, cache results, and retry rate-limit errors (HTTP 403/429) with exponential backoff.

• **Reliability:** submission (FR-10/11) is all-or-nothing per file, with a visible per-file status and retry.

• **Accessibility:** WCAG 2.1 AA; fully usable by keyboard and screen reader.

• **Browsers:** latest two versions of Chrome, Edge, Firefox and Safari, including mobile Safari and Chrome.

9. Success metrics

Area	Metric	v1 target
Activation	% of new projects with 2+ teammates who accepted within 48 h	60%
Engagement	Comments resolved from inside GroupDrive / all resolved comments	40%
Core value	% of projects that submit through GroupDrive before the deadline	70%
Retention	Leads who create a second project in the same term	35%
Reliability	Submission success rate on first try	99%

10. Milestones

Milestone	Scope
M1 (weeks 1-3)	OAuth with drive.file, project creation, invites, file create/pick, file list (FR-1 to FR-5).
M2 (weeks 4-6)	Comment to-do list and resolve, PDF export and keepForever submission (FR-6, FR-7, FR-10, FR-11).
M3 (weeks 7-8)	Activity feed with change notifications, contribution summary, storage check (FR-8, FR-9, FR-14).
M4 (weeks 9-10)	Deadline lock, instructor view, Google OAuth app verification, beta with 2-3 classes (FR-12, FR-13).

11. Risks and mitigations

Risk	Mitigation
Files created outside GroupDrive are invisible (drive.file).	Prominent "New file" and "Add from Drive" buttons; onboarding tip; periodic reminder.
Contribution counts are misread as grades.	Label as activity data only; show counts, not scores; instructor view is opt-in.
Drive API rate limits during deadline spikes.	Queue exports, back off and retry, cache metadata, request minimal fields.
Webhook channels expire or fail.	Renew channels before expiry; fall back to polling changes.list.
Lead's Drive storage is full.	Check storageQuota (FR-14) before exporting; warn and suggest cleanup.

12. Open questions

• Should projects live in a shared drive (POST /drive/v3/drives) instead of the lead's My Drive? Better ownership, but most student accounts can't create shared drives.

• Should we integrate with an LMS (e.g. Canvas) for the actual submission in a later version?

• Is a contribution summary acceptable to students, or should it be hidden from teammates by default?

Appendix: references

• Companion document: *Google Drive API v3 - Capabilities & Documentation* (Google_Drive_API_v3_Capabilities.pdf).

• Drive API REST reference (v3)

• Drive API overview

• Choosing Drive API scopes